II KthSmallestElement.java

```
import java.util.*;
public class KthSmallestElement {
    public static int findkthSmallest(List<Integer> list, int k)
    {
        Collections.sort(list); // Sort the List
        return list.get(k-1); // Return k-th smallest
    }
    public static void main(String[] args) {
        List<Integer> numbers = Arrays.asList(5, 3, 8, 1, 2);
        int k = 3;
        System.out.println("kth smallest: " + findkthSmallest
                                    (numbers, k));
    }
}
```

2 WordFrequencyMap.java

```
import java.util.*;
public class WordFrequencyMap {
  Public static void main(String[] args) {
  String text = "apple banana apple orange banana apple";
  TreeMap<String, Integer> freqMap = new TreeMap<>();
for (String word : text.split(" ")) {
  freqMap.put(word, freqMap.getOrDefault(word,
                                          +1);
System.out.println(freqMap); // output word
                                  frequencies
  }
}
```

[3] QueueStackUsingPriorityQueue.java

```
import java.util.*;
public class QueueStackUsingPriorityQueue {
  static class PQQueue {
    private PriorityQueue<int[]> queue = new PriorityQueue
        <>(Comparator.comparingInt(a -> a[1]));
    private int time = 0;
    public void offer(int val) {
      queue.offer(new int[] {val, time++});
    }
    public int poll() {
      return queue.poll()[0];
    }
  }
}
static class PQStack {
  private PriorityQueue<int[]> stack = new
      PriorityQueue<>((a, b) -> b[1] - a[1]);
  private int time=0;
public void push(int val) {
  stack.offer(new int[] {val, time++}); }
```

```
public int pop (){
    return stack.poll ()[0];
}
public static void main (String[] args) {
    PQQueue q = new PQQueue();
    q.offer (1); q.offer (2); q.offer(3);
    System.out.println("Queue poll: " + q.poll());
    PQStack s = new PQStack ();
    s.push (1); s.push(2); s.push (3);
    System.out.println("Stack pop: " +s.pop());
}
}
```

4. StudentTreeMap.java

```
import java.util.*;
class Student {
    string name;
    int age;
  public Student (String name, int age) {
   this.name = name; this.age = age;
  }
public String toString() {
   return name + " (" + age + ")";
  }
}
public class StudentTreeMap {
  public static void main (String[] args) {
  TreeMap <Integer, Student> students = new TreeMap
                                        <> ();
  students.put (101, new Student ("Alice", 20));
  students.put (102, new student ("Bob", 22));
```

```
for (Map.Entry<Integer, Student> entry :
        students.entrySet()) {
    System.out.println(entry.getKey() + " => " +
                       entry.getValue());
  }
}
```

5 LinkedListEquality.java

```
import java.util.*;
public class LinkedListEquality {
public static void main (String[] args) {
LinkedList<Integer> list1 = new LinkedList<>
        (Arrays.asList(1, 2, 3));
LinkedList<Integer> list2 = new LinkedList<>
        (Arrays.asList(1, 2, 3));
System.out.println("Lists equal? " + list1.equals(list2));
}
}
```

6 EmployeeDepartmentMap.java

```
import java.util.*;
public class EmployeeDepartmentMap {
 public static void main (String[] args) {
 HashMap <Integer, String> employeeMap = new
                                HashMap <> ();

employeeMap.put (1001, "HR");
employeeMap.put (1002, "IT");
employeeMap.put (1003, "Finance");

for (Map.Entry <Integer, String> entry : employeeMap.
                                entrySet ()) {
  System.out.println ("ID: " + entry.getKey () + " => Dept:
                +entry.getValue ());
    }
  }
}
```